

Data Analysis Workflow Library

Expert Workflow Library

Ultimate AI Research Toolkit 2026 - Premium Edition
Designed for PhD students, professors, graduate researchers, engineers, and data scientists.

[Idea]	Research prompts and ideation systems
[Review]	Literature synthesis and gap detection
[Write]	Paper writing and revision workflows
[Plan]	Publication and productivity operations

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Introduction

This Data Analysis collection provides 100 expert research workflows engineered for high-rigor academic and technical delivery. Each workflow includes practical fields to help users move from strategy to publication-ready output.

Instructions

- Select workflows by category and difficulty to match project maturity.
- Replace variables in each workflow with your discipline, data, and resource constraints.
- Use Expected Output as the acceptance criteria for quality control.
- Store outputs in versioned folders to preserve decision history.

Workflow Library

1. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - practical deployment framing

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize rigor and produce publication-grade decisions for Model Diagnostics.

Prompt: Rewrite the methodology around Model Diagnostics for model debugging cycle before resubmission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for IJCAI. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=benchmarking specialist. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Strategy

2. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - method comparison clarity

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize reproducibility and produce publication-grade decisions for Feature Interpretation.

Prompt: Write a Nature-style abstract for a study on Feature Interpretation in deployment drift investigation. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=theory-oriented reviewer. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Analysis

3. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - editorial risk reduction

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize clarity and produce publication-grade decisions for Error Analysis.

Prompt: Generate a figure strategy for Error Analysis with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by IEEE Access. Add one caption rewrite for non-expert readers. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=methodology purist. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Writing

4. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - venue-specific optimization

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize statistical power and produce publication-grade decisions for Data Profiling.

Prompt: Build an ethics checklist and mitigation plan for Data Profiling in post-experiment analysis sprint. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for KDD review. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=interdisciplinary committee member. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Publication

5. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - decision traceability

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize transferability and produce publication-grade decisions for Model Diagnostics.

Prompt: Design a publication strategy for Model Diagnostics: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for IJCAI. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=ethics compliance reviewer. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Methodology

6. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - citation-strength positioning

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize ethical compliance and produce publication-grade decisions for Feature Interpretation.

Prompt: Assume Reviewer #2 challenges Feature Interpretation for PLOS ONE. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=reproducibility auditor. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Execution

7. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - defense-ready reasoning

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize computational efficiency and produce publication-grade decisions for Error Analysis.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Error Analysis. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for IEEE Access. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=applied domain expert. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Compliance

8. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - high-confidence statistical reporting

Objective: Select robust statistical strategy and interpretation logic. Prioritize clinical relevance and produce publication-grade decisions for Data Profiling.

Prompt: Design a statistical analysis workflow for Data Profiling with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for KDD. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=systems performance reviewer. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Reasoning

9. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - evidence-first writing

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize publication readiness and produce publication-grade decisions for Model Diagnostics.

Prompt: Analyze a draft on Model Diagnostics and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to IJCAI. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset;

preferred method=counterfactual evaluation approach; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=industry impact reviewer. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Analysis

10. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - hypothesis stress testing

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize novelty and produce publication-grade decisions for Feature Interpretation.

Prompt: Interpret results for Feature Interpretation using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in PLOS ONE. Add one paragraph that preempts overclaiming. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=editor focused on clarity. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Writing

11. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - results-to-insight translation

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize rigor and produce publication-grade decisions for Error Analysis.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Error Analysis and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for IEEE Access. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=skeptical statistician. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Publication

12. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - major-revision readiness

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize reproducibility and produce publication-grade decisions for Data Profiling.

Prompt: Generate a publication-ready section for Data Profiling tuned for KDD. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=benchmarking specialist. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Methodology

13. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - replication-grade documentation

Objective: Pressure-test experimental design before expensive execution. Prioritize clarity and produce publication-grade decisions for Model Diagnostics.

Prompt: Audit the experimental design for Model Diagnostics. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for IJCAI reviewers. Include a stop/go decision protocol. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=theory-oriented reviewer. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Execution

14. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - reviewer-resilient argumentation

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize statistical power and produce publication-grade decisions for Feature Interpretation.

Prompt: Create a LaTeX writing workflow for Feature Interpretation: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for PLOS ONE submission checks. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=methodology purist. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Compliance

15. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - ethics-by-design reporting

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize transferability and produce publication-grade decisions for Error Analysis.

Prompt: Simulate a high-pressure Q&A; on Error Analysis. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=interdisciplinary committee member. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Reasoning

16. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - limitations-first transparency

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize ethical compliance and produce publication-grade decisions for Data Profiling.

Prompt: Act as a senior research editor. Build a contribution architecture for Data Profiling in post-experiment analysis sprint targeting KDD. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=ethics compliance reviewer. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Rebuttal

17. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - cross-domain readability

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize computational efficiency and produce publication-grade decisions for Model Diagnostics.

Prompt: Rewrite the methodology around Model Diagnostics for model debugging cycle before resubmission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for IJCAI. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=reproducibility auditor. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Validation

18. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - impact-focused communication

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize clinical relevance and produce publication-grade decisions for Feature Interpretation.

Prompt: Write a Nature-style abstract for a study on Feature Interpretation in deployment drift investigation. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=applied domain expert. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Publication

19. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - publishable novelty framing

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize publication readiness and produce publication-grade decisions for Error Analysis.

Prompt: Generate a figure strategy for Error Analysis with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by IEEE Access. Add one caption rewrite for non-expert readers. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=systems performance reviewer. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Methodology

20. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - practical deployment framing

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize novelty and produce publication-grade decisions for Data Profiling.

Prompt: Build an ethics checklist and mitigation plan for Data Profiling in post-experiment analysis sprint. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for KDD review. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=industry impact reviewer. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Execution

21. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - method comparison clarity

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize rigor and produce publication-grade decisions for Model Diagnostics.

Prompt: Design a publication strategy for Model Diagnostics: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for IJCAI. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=editor focused on clarity. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Compliance

22. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - editorial risk reduction

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize reproducibility and produce publication-grade decisions for Feature Interpretation.

Prompt: Assume Reviewer #2 challenges Feature Interpretation for PLOS ONE. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile:

discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=skeptical statistician. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Reasoning

23. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - venue-specific optimization

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize clarity and produce publication-grade decisions for Error Analysis.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Error Analysis. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for IEEE Access. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=benchmarking specialist. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Rebuttal

24. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - decision traceability

Objective: Select robust statistical strategy and interpretation logic. Prioritize statistical power and produce publication-grade decisions for Data Profiling.

Prompt: Design a statistical analysis workflow for Data Profiling with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for KDD. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}.

Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=theory-oriented reviewer. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Validation

25. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - citation-strength positioning

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize transferability and produce publication-grade decisions for Model Diagnostics.

Prompt: Analyze a draft on Model Diagnostics and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to IJCAI. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=methodology purist. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Optimization

26. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - defense-ready reasoning

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize ethical compliance and produce publication-grade decisions for Feature Interpretation.

Prompt: Interpret results for Feature Interpretation using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in PLOS ONE. Add one paragraph that preempts overclaiming. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=interdisciplinary committee member. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Planning

27. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - high-confidence statistical reporting

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize computational efficiency and produce publication-grade decisions for Error Analysis.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Error Analysis and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for IEEE Access. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=ethics compliance reviewer. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Execution

28. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - evidence-first writing

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize clinical relevance and produce publication-grade decisions for Data Profiling.

Prompt: Generate a publication-ready section for Data Profiling tuned for KDD. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=reproducibility auditor. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Compliance

29. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - hypothesis stress testing

Objective: Pressure-test experimental design before expensive execution. Prioritize publication readiness and produce publication-grade decisions for Model Diagnostics.

Prompt: Audit the experimental design for Model Diagnostics. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for IJCAI reviewers. Include a stop/go decision protocol. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=applied domain expert. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Reasoning

30. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - results-to-insight translation

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize novelty and produce publication-grade decisions for Feature Interpretation.

Prompt: Create a LaTeX writing workflow for Feature Interpretation: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for PLOS ONE submission checks. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=systems performance reviewer. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Rebuttal

31. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - major-revision readiness

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize rigor and produce publication-grade decisions for Error Analysis.

Prompt: Simulate a high-pressure Q&A; on Error Analysis. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=industry impact reviewer. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Validation

32. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - replication-grade documentation

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize reproducibility and produce publication-grade decisions for Data Profiling.

Prompt: Act as a senior research editor. Build a contribution architecture for Data Profiling in post-experiment analysis sprint targeting KDD. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=editor focused on clarity. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Optimization

33. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - reviewer-resilient argumentation

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize clarity and produce publication-grade decisions for Model Diagnostics.

Prompt: Rewrite the methodology around Model Diagnostics for model debugging cycle before resubmission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for IJCAI. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=skeptical statistician. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Planning

34. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - ethics-by-design reporting

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize statistical power and produce publication-grade decisions for Feature Interpretation.

Prompt: Write a Nature-style abstract for a study on Feature Interpretation in deployment drift investigation. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=benchmarking specialist. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Strategy

35. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - limitations-first transparency

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize transferability and produce publication-grade decisions for Error Analysis.

Prompt: Generate a figure strategy for Error Analysis with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by IEEE Access.

Add one caption rewrite for non-expert readers. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=theory-oriented reviewer. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Analysis

36. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - cross-domain readability

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize ethical compliance and produce publication-grade decisions for Data Profiling.

Prompt: Build an ethics checklist and mitigation plan for Data Profiling in post-experiment analysis sprint. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for KDD review. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=methodology purist. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Reasoning

37. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - impact-focused communication

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize computational efficiency and produce publication-grade decisions for Model Diagnostics.

Prompt: Design a publication strategy for Model Diagnostics: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for IJCAI. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=interdisciplinary committee member. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Rebuttal

38. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - publishable novelty framing

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize clinical relevance and produce publication-grade decisions for Feature Interpretation.

Prompt: Assume Reviewer #2 challenges Feature Interpretation for PLOS ONE. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=ethics compliance reviewer. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Validation

39. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - practical deployment framing

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize publication readiness and produce publication-grade decisions for Error Analysis.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Error Analysis. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for IEEE Access. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=reproducibility auditor.

Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Optimization

40. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - method comparison clarity

Objective: Select robust statistical strategy and interpretation logic. Prioritize novelty and produce publication-grade decisions for Data Profiling.

Prompt: Design a statistical analysis workflow for Data Profiling with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for KDD. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=applied domain expert. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Planning

41. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - editorial risk reduction

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize rigor and produce publication-grade decisions for Model Diagnostics.

Prompt: Analyze a draft on Model Diagnostics and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to IJCAI. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=systems performance reviewer. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Strategy

42. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - venue-specific optimization

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize reproducibility and produce publication-grade decisions for Feature Interpretation.

Prompt: Interpret results for Feature Interpretation using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in PLOS ONE. Add one paragraph that preempts overclaiming. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=industry impact reviewer. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Analysis

43. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - decision traceability

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize clarity and produce publication-grade decisions for Error Analysis.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Error Analysis and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for IEEE Access. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=editor focused on clarity. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Writing

44. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - citation-strength positioning

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize statistical power and produce publication-grade decisions for Data Profiling.

Prompt: Generate a publication-ready section for Data Profiling tuned for KDD. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=skeptical statistician. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Publication

45. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - defense-ready reasoning

Objective: Pressure-test experimental design before expensive execution. Prioritize transferability and produce publication-grade decisions for Model Diagnostics.

Prompt: Audit the experimental design for Model Diagnostics. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for IJCAI reviewers. Include a stop/go decision protocol. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=benchmarking specialist. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Validation

46. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - high-confidence statistical reporting

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize ethical compliance and produce publication-grade decisions for Feature Interpretation.

Prompt: Create a LaTeX writing workflow for Feature Interpretation: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for PLOS ONE submission checks. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=theory-oriented reviewer. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Optimization

47. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - evidence-first writing

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize computational efficiency and produce publication-grade decisions for Error Analysis.

Prompt: Simulate a high-pressure Q&A; on Error Analysis. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=methodology purist. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Planning

48. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - hypothesis stress testing

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize clinical relevance and produce publication-grade decisions for Data Profiling.

Prompt: Act as a senior research editor. Build a contribution architecture for Data Profiling in post-experiment analysis sprint targeting KDD. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training

setup; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=interdisciplinary committee member. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Strategy

49. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - results-to-insight translation

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize publication readiness and produce publication-grade decisions for Model Diagnostics.

Prompt: Rewrite the methodology around Model Diagnostics for model debugging cycle before resubmission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for IJCAI. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=ethics compliance reviewer. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Analysis

50. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - major-revision readiness

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize novelty and produce publication-grade decisions for Feature Interpretation.

Prompt: Write a Nature-style abstract for a study on Feature Interpretation in deployment drift investigation. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=reproducibility auditor. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Writing

51. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - replication-grade documentation

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize rigor and produce publication-grade decisions for Error Analysis.

Prompt: Generate a figure strategy for Error Analysis with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by IEEE Access. Add one caption rewrite for non-expert readers. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=applied domain expert. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Publication

52. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - reviewer-resilient argumentation

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize reproducibility and produce publication-grade decisions for Data Profiling.

Prompt: Build an ethics checklist and mitigation plan for Data Profiling in post-experiment analysis sprint. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for KDD review. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=systems performance reviewer.

Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Methodology

53. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - ethics-by-design reporting

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize clarity and produce publication-grade decisions for Model Diagnostics.

Prompt: Design a publication strategy for Model Diagnostics: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for IJCAI. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=industry impact reviewer. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Execution

54. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - limitations-first transparency

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize statistical power and produce publication-grade decisions for Feature Interpretation.

Prompt: Assume Reviewer #2 challenges Feature Interpretation for PLOS ONE. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=editor focused on clarity. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor

focused on clarity.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Planning

55. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - cross-domain readability

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize transferability and produce publication-grade decisions for Error Analysis.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Error Analysis. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for IEEE Access. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=skeptical statistician. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Strategy

56. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - impact-focused communication

Objective: Select robust statistical strategy and interpretation logic. Prioritize ethical compliance and produce publication-grade decisions for Data Profiling.

Prompt: Design a statistical analysis workflow for Data Profiling with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for KDD. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=benchmarking specialist. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Analysis

57. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - publishable novelty framing

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize computational efficiency and produce publication-grade decisions for Model Diagnostics.

Prompt: Analyze a draft on Model Diagnostics and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to IJCAI. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=theory-oriented reviewer. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Writing

58. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - practical deployment framing

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize clinical relevance and produce publication-grade decisions for Feature Interpretation.

Prompt: Interpret results for Feature Interpretation using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in PLOS ONE. Add one paragraph that preempts overclaiming. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=methodology purist. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Publication

59. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - method comparison clarity

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize publication readiness and produce publication-grade decisions for Error Analysis.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Error Analysis and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for IEEE Access. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=interdisciplinary committee member. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Methodology

60. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - editorial risk reduction

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize novelty and produce publication-grade decisions for Data Profiling.

Prompt: Generate a publication-ready section for Data Profiling tuned for KDD. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=ethics compliance reviewer. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Execution

61. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - venue-specific optimization

Objective: Pressure-test experimental design before expensive execution. Prioritize rigor and produce publication-grade decisions for Model Diagnostics.

Prompt: Audit the experimental design for Model Diagnostics. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for IJCAI reviewers. Include a stop/go decision protocol. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=reproducibility auditor. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Compliance

62. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - decision traceability

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize reproducibility and produce publication-grade decisions for Feature Interpretation.

Prompt: Create a LaTeX writing workflow for Feature Interpretation: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for PLOS ONE submission checks. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=applied domain expert. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Reasoning

63. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - citation-strength positioning

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize clarity and produce publication-grade decisions for Error Analysis.

Prompt: Simulate a high-pressure Q&A; on Error Analysis. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution

checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=systems performance reviewer. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Analysis

64. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - defense-ready reasoning

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize statistical power and produce publication-grade decisions for Data Profiling.

Prompt: Act as a senior research editor. Build a contribution architecture for Data Profiling in post-experiment analysis sprint targeting KDD. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=industry impact reviewer. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Writing

65. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - high-confidence statistical reporting

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize transferability and produce publication-grade decisions for Model Diagnostics.

Prompt: Rewrite the methodology around Model Diagnostics for model debugging cycle before resubmission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for IJCAI. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=editor focused on clarity. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Publication

66. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - evidence-first writing

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize ethical compliance and produce publication-grade decisions for Feature Interpretation.

Prompt: Write a Nature-style abstract for a study on Feature Interpretation in deployment drift investigation. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=skeptical statistician. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Methodology

67. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - hypothesis stress testing

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize computational efficiency and produce publication-grade decisions for Error Analysis.

Prompt: Generate a figure strategy for Error Analysis with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by IEEE Access. Add one caption rewrite for non-expert readers. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=benchmarking specialist. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Execution

68. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - results-to-insight translation

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize clinical relevance and produce publication-grade decisions for Data Profiling.

Prompt: Build an ethics checklist and mitigation plan for Data Profiling in post-experiment analysis sprint. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for KDD review. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=theory-oriented reviewer. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Compliance

69. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - major-revision readiness

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize publication readiness and produce publication-grade decisions for Model Diagnostics.

Prompt: Design a publication strategy for Model Diagnostics: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for IJCAI. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=methodology purist. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Reasoning

70. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - replication-grade documentation

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize novelty and produce publication-grade decisions for Feature Interpretation.

Prompt: Assume Reviewer #2 challenges Feature Interpretation for PLOS ONE. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=interdisciplinary committee member. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Rebuttal

71. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - reviewer-resilient argumentation

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize rigor and produce publication-grade decisions for Error Analysis.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Error Analysis. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for IEEE Access. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=ethics compliance reviewer. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Validation

72. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - ethics-by-design reporting

Objective: Select robust statistical strategy and interpretation logic. Prioritize reproducibility and produce publication-grade decisions for Data Profiling.

Prompt: Design a statistical analysis workflow for Data Profiling with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for KDD. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=reproducibility auditor. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Publication

73. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - limitations-first transparency

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize clarity and produce publication-grade decisions for Model Diagnostics.

Prompt: Analyze a draft on Model Diagnostics and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to IJCAI. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=applied domain expert. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Methodology

74. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - cross-domain readability

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize statistical power and produce publication-grade decisions for Feature Interpretation.

Prompt: Interpret results for Feature Interpretation using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in PLOS ONE. Add one paragraph that preempts overclaiming. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=systems performance reviewer. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Execution

75. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - impact-focused communication

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize transferability and produce publication-grade decisions for Error Analysis.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Error Analysis and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for IEEE Access. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=industry impact reviewer. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Compliance

76. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - publishable novelty framing

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize ethical compliance and produce publication-grade decisions for Data Profiling.

Prompt: Generate a publication-ready section for Data Profiling tuned for KDD. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=editor focused on clarity. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Reasoning

77. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - practical deployment framing - Variant 077

Objective: Pressure-test experimental design before expensive execution. Prioritize computational efficiency and produce publication-grade decisions for Model Diagnostics.

Prompt: Audit the experimental design for Model Diagnostics. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for IJCAI reviewers. Include a stop/go decision protocol. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=skeptical statistician. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Rebuttal

78. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - method comparison clarity - Variant 078

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize clinical relevance and produce publication-grade decisions for Feature Interpretation.

Prompt: Create a LaTeX writing workflow for Feature Interpretation: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for PLOS ONE submission checks. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer

disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=benchmarking specialist. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Validation

79. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - editorial risk reduction - Variant 079

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize publication readiness and produce publication-grade decisions for Error Analysis.

Prompt: Simulate a high-pressure Q&A; on Error Analysis. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=theory-oriented reviewer. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Optimization

80. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - venue-specific optimization - Variant 080

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize novelty and produce publication-grade decisions for Data Profiling.

Prompt: Act as a senior research editor. Build a contribution architecture for Data Profiling in post-experiment analysis sprint targeting KDD. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=methodology purist. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Planning

81. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - decision traceability - Variant 081

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize rigor and produce publication-grade decisions for Model Diagnostics.

Prompt: Rewrite the methodology around Model Diagnostics for model debugging cycle before resubmission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for IJCAI. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=interdisciplinary committee member. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Execution

82. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - citation-strength positioning - Variant 082

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize reproducibility and produce publication-grade decisions for Feature Interpretation.

Prompt: Write a Nature-style abstract for a study on Feature Interpretation in deployment drift investigation. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=ethics compliance reviewer. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Compliance

83. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - defense-ready reasoning - Variant 083

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize clarity and produce publication-grade decisions for Error Analysis.

Prompt: Generate a figure strategy for Error Analysis with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by IEEE Access. Add one caption rewrite for non-expert readers. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=reproducibility auditor. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Reasoning

84. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - high-confidence statistical reporting - Variant 084

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize statistical power and produce publication-grade decisions for Data Profiling.

Prompt: Build an ethics checklist and mitigation plan for Data Profiling in post-experiment analysis sprint. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for KDD review. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=applied domain expert. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Rebuttal

85. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - evidence-first writing - Variant 085

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize transferability and produce publication-grade decisions for Model Diagnostics.

Prompt: Design a publication strategy for Model Diagnostics: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for IJCAI. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=systems performance reviewer. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Validation

86. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - hypothesis stress testing - Variant 086

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize ethical compliance and produce publication-grade decisions for Feature Interpretation.

Prompt: Assume Reviewer #2 challenges Feature Interpretation for PLOS ONE. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=industry impact reviewer. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Optimization

87. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - results-to-insight translation - Variant 087

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize computational efficiency and produce publication-grade decisions for Error Analysis.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Error Analysis. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for IEEE Access. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=editor focused on clarity. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Planning

88. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - major-revision readiness - Variant 088

Objective: Select robust statistical strategy and interpretation logic. Prioritize clinical relevance and produce publication-grade decisions for Data Profiling.

Prompt: Design a statistical analysis workflow for Data Profiling with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for KDD. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=skeptical statistician. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Strategy

89. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - replication-grade documentation -

Variant 089

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize publication readiness and produce publication-grade decisions for Model Diagnostics.

Prompt: Analyze a draft on Model Diagnostics and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to IJCAI. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=benchmarking specialist. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Analysis

90. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - reviewer-resilient argumentation - Variant 090

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize novelty and produce publication-grade decisions for Feature Interpretation.

Prompt: Interpret results for Feature Interpretation using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in PLOS ONE. Add one paragraph that preempts overclaiming. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=theory-oriented reviewer. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Reasoning

91. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - ethics-by-design reporting - Variant 091

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize rigor and produce publication-grade decisions for Error Analysis.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Error Analysis and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for IEEE Access. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=methodology purist. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Rebuttal

92. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - limitations-first transparency - Variant 092

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize reproducibility and produce publication-grade decisions for Data Profiling.

Prompt: Generate a publication-ready section for Data Profiling tuned for KDD. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=interdisciplinary committee member. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Validation

93. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - cross-domain readability - Variant 093

Objective: Pressure-test experimental design before expensive execution. Prioritize clarity and produce publication-grade decisions for Model Diagnostics.

Prompt: Audit the experimental design for Model Diagnostics. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for IJCAI reviewers. Include a stop/go decision protocol. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=ethics compliance reviewer. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from an ethics compliance reviewer.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Optimization

94. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - impact-focused communication - Variant 094

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize statistical power and produce publication-grade decisions for Feature Interpretation.

Prompt: Create a LaTeX writing workflow for Feature Interpretation: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for PLOS ONE submission checks. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=reproducibility auditor. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Planning

95. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - publishable novelty framing - Variant 095

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize transferability and produce publication-grade decisions for Error Analysis.

Prompt: Simulate a high-pressure Q&A; on Error Analysis. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=applied domain expert. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Strategy

96. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - practical deployment framing - Variant 096

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize ethical compliance and produce publication-grade decisions for Data Profiling.

Prompt: Act as a senior research editor. Build a contribution architecture for Data Profiling in post-experiment analysis sprint targeting KDD. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=systems performance reviewer. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Analysis

97. Diagnose Failure Modes with Evidence: diagnostic reporting system for model debugging cycle before resubmission (IJCAI) - method comparison clarity - Variant 097

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize computational efficiency and produce publication-grade decisions for Model Diagnostics.

Prompt: Rewrite the methodology around Model Diagnostics for model debugging cycle before resubmission so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for IJCAI. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IJCAI; reviewer persona=industry impact reviewer. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Model Diagnostics.

Difficulty: Beginner

Category: Writing

98. Build a High-Rigor Error Analysis: feature-impact investigation for deployment drift investigation (PLOS ONE) - editorial risk reduction - Variant 098

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize clinical relevance and produce publication-grade decisions for Feature Interpretation.

Prompt: Write a Nature-style abstract for a study on Feature Interpretation in deployment drift investigation. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PLOS ONE; reviewer persona=editor focused on clarity. Example outcome: feature impact memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready feature impact memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Feature Interpretation.

Difficulty: Intermediate

Category: Publication

99. Convert Metrics into Insight: result-explanation workflow for cross-cohort transfer study (IEEE Access) - venue-specific optimization - Variant 099

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize publication readiness and produce publication-grade decisions for Error Analysis.

Prompt: Generate a figure strategy for Error Analysis with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by IEEE Access. Add one caption rewrite for non-expert readers. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Access; reviewer persona=skeptical statistician. Example outcome: analysis report with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready analysis report that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Error Analysis.

Difficulty: Advanced

Category: Validation

100. Explain Results Scientifically: error taxonomy workflow for post-experiment analysis sprint (KDD) - decision traceability - Variant 100

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize novelty and produce publication-grade decisions for Data Profiling.

Prompt: Build an ethics checklist and mitigation plan for Data Profiling in post-experiment analysis sprint. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for KDD review. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=KDD; reviewer persona=benchmarking specialist. Example outcome: diagnostic dashboard with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready diagnostic dashboard that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Data Profiling.

Difficulty: Expert

Category: Optimization

Summary

Applying these 100 workflows systematically improves decision quality, writing clarity, and publication readiness across research cycles.